# Trajectory-exact constrained decoding for masked diffusion models, for any tractable constraint

Working notes

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**Abstract**

Masked diffusion language models fill in a sequence over several steps, revising their predictions as positions are revealed. Constrained decoders that enforce a constraint one step at a time ignore those revisions and sample from the wrong distribution. TWISTER fixes this for constraints given by deterministic finite automata. These notes separate the fix from the automaton: any structure that can compute the probability of a constraint under a product distribution, and sample from that distribution restricted to the constraint, can take the automaton's place: grammar charts, factor graphs and determinants, with ambiguous grammars allowed. We give the general theorem, the practical failure mode that shows up when the method meets a real model, three remedies with guarantees, experiments on a text diffusion model and a synthetic testbed, and the state of an application to mRNA sequence design.

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# 1 What this is, and why

## 1.1 The problem

A masked diffusion language model [19] builds a sequence of length $L$ from a row of mask symbols over $T$ steps. At each step a *denoiser* reads the current partial sequence and predicts a distribution over the vocabulary at every masked position; a *scheduler* picks which positions to reveal; those positions are filled independently from the predictions. We call this the *native sampler* and write $p$ for the distribution it induces.

Two facts about it drive everything below. Within a step, the predictions are a product distribution across positions. Across steps, they change, because every denoiser call conditions on what has been revealed so far.

Now fix a set $C$ of acceptable outputs and ask for $p(x \mid x \in C)$: the model's own distribution, restricted. Rejection sampling is exact and useless when $p(C)$ is small. The cheap alternatives all distort the distribution. Token masking forbids tokens that make $C$ unreachable, which ignores how much probability stays inside $C$ after each choice. *Step-exact decoding* [5] does better, restricting

the current product distribution to $C$ and sampling from that, and is still wrong, for a reason best seen concretely.

**Example 1.1** (Why one-step-at-a-time fails)**.** Let $L = 2$, $T = 2$, reveal position 1 then position 2, and let $C = \{(a, a)\}$: the two tokens must agree. Write $p^1(a)$ for the prediction at position 1 from the all-mask state, $p^2(a \mid \varnothing)$ for the prediction at position 2 from that same state, and $p^2(a \mid a)$ for the prediction at position 2 after position 1 was revealed as $a$. The native conditional is

$$p(x = (a, a) \mid C) \;\propto\; p^1(a)\, p^2(a \mid a).$$

Step-exact decoding draws position 1 from the restricted product, giving $a$ with probability $\propto p^1(a)\, p^2(a \mid \varnothing)$, and the second step is then forced. The two laws agree only when $p^2(a \mid a)/p^2(a \mid \varnothing)$ is constant in $a$: only when revealing position 1 tells the model nothing about position 2. For a real model it always tells it something.

That ratio is the whole story. It compares the constraint's probability under the *updated* predictions with its probability under the *old* ones, and it is exactly the importance weight that repairs the bias. TWISTER [22] applies it at every step, runs a population of particles, and proves the result targets the native conditional exactly, for constraints given by deterministic finite automata.

### 1.2 The claim

**Main claim.**

The trajectory-level correction does not depend on the constraint being an automaton. It needs a structure that (i) computes the constraint's probability under a product distribution, (ii) samples from that product distribution restricted to the constraint, and (iii) tests membership exactly at the end. We call such a structure a *route*. Automata, grammar charts, factor graphs and determinants are all routes, and each covers constraints the others cannot handle in polynomial time. The route need not even be unambiguous: a grammar that counts a string once per parse tree still gives the right answer.

The architecture follows the claim. A *route* handles everything inside one step. A *trajectory layer*, twisted sequential Monte Carlo, handles what happens across steps and is agnostic to which route sits underneath. Theorem 4.2 is the statement that this separation is sound.

### 1.3 What these notes add

Three things beyond the route-generic theorem.

**When the correction is unnecessary.** Proposition 5.2 identifies the degenerate case: if the denoiser's predictions do not move as positions are revealed, the frozen twist is already the ideal one, and step-exact decoding is already trajectory-exact. The method's value on a given model is therefore governed by one measurable quantity, which we call re-conditioning strength. This is both a theoretical boundary and a cheap pre-test.

**What breaks on a real model.** The Monte Carlo estimate of $p(C)$ is unbiased at every particle count. But when the denoiser badly understates the constraint's chances at some intermediate step, adaptive resampling discards particles that would have recovered, and a substantial share of the estimate's expectation ends up in events too rare to ever observe. On a text diffusion model this cost us 7% of $p(C)$. Section 6 characterises this *hidden mass* exactly and gives three remedies that keep the estimate unbiased.

**A negative result about ambiguity.** Ambiguous grammars are fine without resampling and get worse with it, for a reason that admits no exact fix (Section 7).

Each result is tagged [background] (known, cited), [ours] (proved here) or [proposed] (a direction). Section 12 lists them all. Proofs longer than a few lines are in Appendix A.

## 2 Setup

**Tokens and classes.** Let $V$ be the vocabulary and $\text{M} \notin V$ the mask. Constraints are stated over a small alphabet $\Sigma$ of token *classes* through a map $\phi : V \to \Sigma$ (digit, opening bracket, codon). We write $C \subseteq \Sigma^L$ and say a token sequence $x$ satisfies $C$ if $\phi(x) \in C$. For a partial sequence $x \in (V \cup \{\text{M}\})^L$, $\mathcal{M}(x)$ is its set of masked positions.

**Native kernel.** One native step from $x$ to $x'$ has probability

$$\kappa_{t+1}(x' \mid x) = s_{t+1}(r \mid x) \prod_{i \in r} p_\theta^i(x'_i \mid x), \tag{1}$$

where $r \subseteq \mathcal{M}(x)$ is the revealed set, $s_{t+1}$ the scheduler, and $p_\theta^i$ the denoiser's prediction. The native path law is $p(x_{0:T}) = \prod_t \kappa_{t+1}$, starting from $x_0 = \text{M}^L$.

**Reveal schedules.** The scheduler is part of the process, not an implementation detail, and two choices are in common use. The *fixed* schedule reveals exactly $L/T$ positions per step, uniformly among the masked ones; this is TWISTER's. The *independent* schedule reveals each masked position with probability $1/(T-t)$, which is the MDLM reverse step [19] under the log-linear noise schedule and leaves a random number of reveals per step. Everything below holds for either, since $s_{t+1}$ cancels in (4). But it must be the *same* scheduler in the proposal and in the native law being conditioned. The two laws are genuinely different: on identical models and constraints, $p(C)$ differs between them by 4 to 10 standard errors at $n = 3 \times 10^5$ samples. mRNAutilus uses the independent schedule, so our mRNA runs do too.

**Target.** The *Doob path law* $p^\star(x_{0:T}) \propto p(x_{0:T})\mathbf{1}_C(x_T)$ has terminal marginal $p(\cdot \mid C)$, and is a Markov chain with kernel $\kappa_{t+1}h_{t+1}/h_t$, where

$$h_t(x) = \Pr_p(X_T \in C \mid X_t = x)$$

is the *Doob function*. It depends on the denoiser's outputs at every future state, so it cannot be computed. Everything in these notes is about what to use in its place. We write $h_0 = p(C)$.

**Frozen weights and the clamped sum.** For a partial sequence $x$ the class weights are $w_i(\sigma \mid x) = \sum_{v:\phi(v)=\sigma} p_\theta^i(v \mid x)$ at masked positions, and $w_i(\sigma \mid x) = [\sigma = \phi(x_i)]$ at revealed ones. The *clamped sum* of a partial sequence $x'$ under the predictions made at $x$ is

$$Z(x'; x) = \sum_{y \in C} \prod_{i \in \mathcal{M}(x')} w_i(y_i \mid x) \prod_{i \notin \mathcal{M}(x')} [y_i = \phi(x'_i)]. \tag{2}$$

Then $\eta_t(x) := Z(x; x)$ is the probability that $C$ holds if every masked position is filled independently from the current predictions. TWISTER uses $\eta_t$ in place of $h_t$; we call it the *frozen twist*, since it treats the current predictions as final.

**Twisted SMC in one paragraph.** Standard material; [6] covers all of it. A *twist* is a sequence $\psi_0, \ldots, \psi_T$ with $\psi_T = \mathbf{1}_C$, defining intermediate targets $\gamma_t = p(x_{0:t})\psi_t(x_t)$ and *potentials*

$$G_{t+1}(x_t, x_{t+1}) = \frac{\kappa_{t+1}(x_{t+1} \mid x_t)\, \psi_{t+1}(x_{t+1})}{M_{t+1}(x_{t+1} \mid x_t)\, \psi_t(x_t)} \tag{3}$$

for a proposal $M$. The product $\psi_0 \prod_t G_{t+1}$ telescopes to $p\mathbf{1}_C/q$, so the *final* weight does not depend on the twist; the twist changes only the intermediate weights, and hence which particles survive resampling. The ideal twist is $h_t$. With $K$ particles, resampling fires when the effective sample size $\mathrm{ESS} = (\sum_k W^k)^2 / \sum_k (W^k)^2$ drops below a threshold, and $\widehat{Z} = \prod_{\text{resamplings}} \left(\frac{1}{K}\sum_k W^k\right) \times \left(\frac{1}{K}\sum_k W^k\right)_{\text{final}}$ is unbiased for $h_0$ at every $K$.

# 3 Routes

**Definition 3.1** (Route)**.** A *route* for $C$ provides three operations on weights $w \geq 0$:

- $\textsc{Mass}(w) = \sum_y d(y) \prod_i w_{i,y_i}$, for a fixed $d \geq 0$ with $\operatorname{supp} d \supseteq C$;
- $\textsc{Sample}(w)$, an exact draw from $y \propto d(y) \prod_i w_{i,y_i}$;
- $\textsc{Accepts}(y) = \mathbf{1}_C(y)$, an exact membership test.

It is *exact* if $d = \mathbf{1}_C$, *derivation-weighted* if $d \in \mathbb{N}$ with $\operatorname{supp} d = C$, and a *superset route* if $\operatorname{supp} d \supsetneq C$.

The condition the trajectory layer relies on is the one that is easy to get wrong: Sample must draw from exactly the measure whose total Mass returns. Here $d(y)$ counts *derivations*: accepting paths of an automaton, parse trees of a grammar. A route is unambiguous when $d \equiv 1$ on $C$.

## 3.1 The four families

Table 1 summarises them; Appendix B gives the recursions, the samplers and their references.

| route | cost of one pass | sampler | covers |
|---|---|---|---|
| automaton | $O(L\lvert\delta\rvert)$ | forward-backward | motifs, counts, regular expressions, bracket depth |
| grammar chart | $O(L^3\lvert G\rvert)$ | top-down from the chart | context-free constraints, RNA pairing |
| factor graph [12] | $O(L\lvert\Sigma\rvert^{\omega+1})$ | elimination then reverse | copy relations, target structures |
| determinant | $O(n^3)$ | Wilson's algorithm | spanning trees, dependency structures |

Table 1: The four route families. $\omega$ is the induced width of the elimination order.

Which route is cheapest depends on the representation, not only the constraint. The same-order copy relation $y_i = y_{i+k}$ is width-1 as a factor graph and needs $|\Sigma|^k$ states as an in-order automaton. A compiler estimates the cost of each candidate from compile-time statistics and picks the cheapest; when none fits the budget, or the family has #P-hard counting, it refuses rather than approximating.

## 3.2 Ambiguity is a feature of the route, not the constraint

The same language can have an unambiguous route and an ambiguous one. Unambiguous Dyck-1 is $S \to (S)S \mid \varepsilon$; ambiguous Dyck-1 is $S \to SS \mid (S) \mid ()$, which generates the same strings but gives a string with several adjacent blocks several parse trees. The distinction matters because of where it shows up in the semiring.

**Proposition 3.2** ([background]). *Let a structure circuit $S$ represent $C$. Evaluated on leaves $w \geq 0$, the Boolean semiring answers completability and max-times answers MAP, both regardless of ambiguity; sum-product returns $\sum_y d_S(y) \prod_i w_{i,y_i}$, which is the constrained probability* if and only if *$d_S \equiv 1$ on $C$.*

*Proof.* A special case of Wang et al. [23], Thm. 4 and Ex. 2: mapping a Boolean circuit into a semiring is tractable when the semiring is idempotent or the circuit is deterministic. The underlying fact is older and more general: the correctness of a sum-product recursion uses only distributivity and the commutation of nested sums, so any commutative semiring gives a valid algorithm — the generalized distributive law of Aji and McEliece [1].

Idempotent addition counts the $d_S(y)$ copies of a string once; ordinary addition counts all of them. So an ambiguous route samples from the derivation-weighted law $\propto d(y) \prod_i w_{i,y_i}$. Dang and Ermon call their NFA variant a "soft proxy" for this reason [5]. One useful corollary: Viterbi on the length-$L$ chart computes exact constrained MAP for *every* context-free grammar, inherently ambiguous ones included, in $O(L^3|G|)$. The surprise of Theorem 4.2 is that sampling, too, survives ambiguity.

### 3.3 The tractability boundary

Step-exact decoding needs the constrained probability and its marginals. Both are queries on the *constraint polynomial* $P_C(w) = \sum_{y \in C} \prod_i w_{i,y_i}$, and their tractability is settled.

**Theorem 3.3** ([background]). *Fix $\Sigma$ and a family $\{C_L\}$, with all bounds polynomial in $L$ and $|\Sigma|$. These are equivalent: (a) $P_{C_L}$ has a polynomial-size arithmetic circuit over $\mathbb{R}$, where subtraction is allowed and the circuit need not be decomposable, deterministic or monotone; (b) a real-RAM algorithm computes $P_{C_L}(w)$ for every $w \geq 0$; (c) a real-RAM algorithm computes all constrained marginals whenever $P_{C_L}(w) > 0$. Given a circuit of size $s$, $P_C$ costs $O(s)$, all $L|\Sigma|$ marginals together cost $O(s)$, and an exact sample costs $O(Ls)$.*

*Proof.* (b)⇒(a) is Broadrick et al. [3], Prop. 5.4. Their virtual-evidence marginalisation is query (b) for $\mathbf{1}_C$ after one-hot encoding; we checked that their division-elimination and polynomial-identity steps go through for $P_C$ with correctness only on the positive orthant. (a)⇒(b) is evaluation; marginals come from one reverse-mode pass. (c)⇒(b) and the sampler are self-reducibility under clamping [11].

Two boundary facts matter for what follows. First, the requirement is *arbitrary* weights: tractable marginals under hard evidence alone do not imply a small circuit [3], and decoding does supply arbitrary weights, because the weights are the model's predictions. Second, decomposable circuits (which automata, charts and bounded-width factor graphs all are) do not cover everything.

**Remark 3.4** (Spanning trees separate the two [background]). Encode a tree on $\{0, \ldots, n\}$ by head pointers and let $C$ be the arborescences rooted at 0. Then $P_C$ is the directed spanning-tree polynomial: polynomial-size via the matrix-tree theorem, but every *monotone* circuit for it has size $2^{\Omega(n)}$ [10]. Smooth decomposable circuits with non-negative parameters are monotone, so no automaton, grammar or bounded-treewidth route is polynomial here, while exact decoding through a determinant is. This is why the determinant route exists.

Below the boundary there is nothing to be done: if a family has no FPRAS it has no polynomial-time almost-uniform sampler either [11], so the compiler's refusal is the correct behaviour rather than a limitation of the implementation.

## 4 The algorithm, and why it is exact

Write $Z_d(x'; x)$ for the clamped sum (2) with $\mathbf{1}_C$ replaced by the route's $d$, and $\eta_t(x) = Z_d(x; x)$.

**Lemma 4.1** (Step-exact proposal [background]). *The kernel $M_{t+1}(x' \mid x) = s_{t+1}(r \mid x) \prod_{i \in r} p_\theta^i(x'_i \mid x)\, Z_d(x'; x)/Z_d(x; x)$ is a probability kernel, sampled by drawing $r$ from the native scheduler, drawing a complete class sequence $y \sim \textsc{Sample}(w(x))$ with revealed positions clamped, drawing tokens within classes at the positions in $r$, and revealing only those.*

*Proof.* TWISTER [22], Prop. 3.6, stated for DFAs. The proof uses only that the sampler draws from the measure whose total is $Z_d(x; x)$, so it carries over to any route.

Combining Lemma 4.1 with (3), the scheduler and the token probabilities cancel:

$$G_{t+1}(x, x') = \frac{Z_d(x; x)}{\psi_t(x)} \cdot \frac{\psi_{t+1}(x')}{Z_d(x'; x)}. \tag{4}$$

With TWISTER's choice $\psi_t = \eta_t$ this is $G_{t+1} = Z_d(x'; x')/Z_d(x'; x)$: the mass of the new partial sequence under the new predictions over its mass under the old ones. In Example 1.1 it is exactly $p^2(a \mid a)/p^2(a \mid \varnothing)$.

**Algorithm 1** Route-generic twisted SMC (one run with $K$ particles)

**Require:** route ($\textsc{Mass}, \textsc{Sample}, \textsc{Accepts}$); denoiser; $T$; $K$; ESS threshold $\tau$; $\epsilon \in [0, 1)$; $\beta \in (0, 1]$

for each $k$: $x^k \leftarrow \text{M}^L$, $\eta^k \leftarrow \textsc{Mass}(w(x^k))$, $\log W^k \leftarrow \beta \log \eta^k$; $\log \widehat{Z} \leftarrow 0$
**for** $t = 0, \ldots, T-1$ **do**
  **for** each particle $k$ **do**
    draw $r \sim s_{t+1}(\cdot \mid x^k)$ and $y \sim \textsc{Sample}(w(x^k))$; draw tokens within classes on $r$; $x' \leftarrow x^k$ revealed on $r$
    $z \leftarrow Z_d(x'; x^k)$ ▷ mass of $x'$ under the old predictions
    **if** $t < T-1$ **then** call the denoiser at $x'$; $\eta' \leftarrow Z_d(x'; x')$; $\psi' \leftarrow (\eta')^\beta$
    **else** $\psi' \leftarrow \textsc{Accepts}(x')$ ▷ exact membership at the last step
    $\log W^k \leftarrow \log W^k + (1-\beta) \log \eta^k + \log \psi' - \log z$; $x^k \leftarrow x'$; $\eta^k \leftarrow \eta'$
  **if** $t < T-1$ and $\text{ESS}(W) < \tau K$ **then**
    $\log \widehat{Z} \leftarrow \log \widehat{Z} + \log \frac{1}{K} \sum_k W^k$; $\tilde{w} \leftarrow W / \sum_k W^k$; $\pi \leftarrow (1-\epsilon)\tilde{w} + \epsilon/K$
    draw ancestors $a_1, \ldots, a_K$ from $\pi$; for each $k$: $x^k \leftarrow x^{a_k}$, $\eta^k \leftarrow \eta^{a_k}$, $W^k \leftarrow \tilde{w}_{a_k}/\pi_{a_k}$
$\log \widehat{Z} \leftarrow \log \widehat{Z} + \log \frac{1}{K} \sum_k W^k$
**return** weighted particles $\{(x^k, W^k)\}$ and $\widehat{Z}$

Algorithm 1 is TWISTER with the automaton replaced by a route, plus two options introduced in Section 6: defensive mixing $\epsilon$ and twist power $\beta$. At $\epsilon = 0, \beta = 1$ it is TWISTER. At $K = 1$ with the weights ignored it is step-exact decoding [5]. Per particle and step it costs one denoiser call, one route sample and two mass evaluations, the second of which is reused next step.

**Theorem 4.2** (Route-generic trajectory exactness [ours]). *Assume the denoiser gives every token positive probability, the route satisfies* $\operatorname{supp} d \supseteq C$ *with ambiguity allowed, and the twists satisfy $\psi_t > 0$ wherever $h_t > 0$ with $\psi_T = \mathbf{1}_C$. Then:*

*(i) $q(x_{0:T})\, W(x_{0:T}) = p(x_{0:T})\, \mathbf{1}_C(x_T)$ for every path;*
*(ii) the weighted particles target the Doob path law and are consistent for $p(\cdot \mid C)$ as $K \to \infty$;*
*(iii) $\mathbb{E}[\widehat{Z}] = h_0$ for every $K \geq 1$, every $\epsilon \in [0, 1)$, and every resampling rule whose decision depends only on the current particles and weights;*

*(iv) the derivation count affects only the variance, since the factor $d(x_T)$ from the last proposal step cancels in the last potential.*

*Proof idea.* By Lemma 4.1 the route's clamped sums appear in both the proposal density and the potential and cancel, leaving a product of twists that telescopes to $p\,\psi_T$. Part (iv) is the reason ambiguity is survivable: at the last step $x_T$ is fully revealed, so $Z_d(x_T; x_{T-1}) = d(x_T)$ appears in the denominator of the final potential and cancels the $d(x_T)$ that the proposal introduced. Full proof in Appendix A.

This separates what each kind of decoding needs. Step-exact decoding needs an exact route (Theorem 3.3). Trajectory-exact decoding needs only a route whose sampler matches its mass, a twist positive on completable states, and an exact membership test at the end. Any NFA or CFG chart meets these in polynomial time per particle. There is no contradiction with the #P-hardness of computing $h_t$ for general CFGs, because SMC never computes $h_t$; it is exact only in the limit.

**Proposition 4.3** (Terminal-twist pitfall [ours])**.** *If an ambiguous route's own value $Z_d(x_T; x_T) = d(x_T)$ is used as the terminal twist in place of $\mathbf{1}_C$, the particles target $\propto p(x_{0:T})\, d(x_T)$, the derivation-weighted law rather than the one asked for.*

*Proof.* Replace $\psi_T = \mathbf{1}_C$ by $\psi_T = d$ in the telescoping product of Theorem 4.2(i).

This is an easy implementation error, since MASS at a fully revealed state looks like a membership test and is not one. The last step must call ACCEPTS.

**Remark 4.4** (Relation to prior work)**.** For unambiguous routes Theorem 4.2 is TWISTER's argument [22] (Thm. 4.2) applied to a different route; their proof uses the DFA only to compute $Z$. Dang et al. [4] observe, for NFAs in autoregressive SMC, that the derivation-weighted distribution is still a valid proposal. Theorem 4.2 combines these for masked diffusion and every route of Section 3. We have not found the terminal-twist pitfall or the analysis of Section 6 stated elsewhere.

### 4.1 The twist is free

The proof uses only two facts: the proposal density is computable, and the twists telescope. So the twist may be chosen independently of the proposal, which is what every remedy in Section 6 exploits.

**Proposition 4.5** (Free intermediate twists [background])**.** *Let $M_{t+1}$ be any kernel with computable density, positive wherever $\kappa_{t+1} h_{t+1} > 0$, and let $\psi_t \geq 0$ be positive where $h_t > 0$ with $\psi_T = \mathbf{1}_C$. Then $qW = p\mathbf{1}_C$, and (ii) and (iii) of Theorem 4.2 hold. The same is true for* random *twists $\psi_t(x, \xi_t)$, provided the realisation $\xi_t$ is reused in the numerator at step $t$ and the denominator at step $t+1$.*

*Proof.* The deterministic case is the standard twisted-SMC construction [6]; [25] uses it for autoregressive language models. For random twists, include $\xi_t$ in the state; each realisation appears once in a numerator and once in a denominator, so the product still telescopes path by path.

### 4.2 A diagnostic identity

**Lemma 4.6** ([ours])**.** *Under the conditions of Theorem 4.2, $\mathbb{E}_q[W] = h_0$ and $\mathbb{E}_p[\mathbf{1}_C(x_T)/W] = \Pr_q(x_T \in C)$, the second expectation taken over native paths. For an exact or derivation-weighted route the right-hand side is 1.*

*Proof.* By Theorem 4.2(i), $\mathbb{E}_p[\mathbf{1}_C/W] = \sum_{\text{paths}} p\,\mathbf{1}_C\,q/(p\,\mathbf{1}_C)$, the $q$-probability of ending in $C$.

The first identity is checked with proposal runs, the second with native samples; together they catch different bugs. Native paths whose weight exceeds every observed proposal weight point to regions the proposal rarely visits. Running Algorithm 1 with resampling switched off alongside the main run separates proposal-and-twist error from resampling error, which is how the problem in the next section was found.

## 5 When the correction is unnecessary

The form of $G$ in (4) says the entire trajectory-level correction is driven by one thing: how much the denoiser's predictions move as positions are revealed. The degenerate case delimits exactly when this machinery earns its cost.

**Definition 5.1** (Trajectory mean-field [ours])**.** A denoiser is *trajectory mean-field* if $p_\theta^i(\cdot \mid x) = p_\theta^i(\cdot \mid \tilde{x})$ for every $i$ and every pair of partial sequences with $\mathcal{M}(x) = \mathcal{M}(\tilde{x})$: the prediction at a masked position depends on which positions are masked, but not on the values revealed at the others.

**Proposition 5.2** (Mean-field collapse [ours])**.** *Let the denoiser be trajectory mean-field, let $\psi_t = \eta_t$ for $t < T$ and $\psi_T = \mathbf{1}_C$, and take an exact route. Then $\eta_t = h_t$ everywhere and $G_{t+1} = 1$ almost surely, so with $K = 1$ and no resampling, $\widehat{Z} = \eta_0 = h_0$ deterministically and the single trajectory is an exact draw from $p(\cdot \mid C)$. Step-exact decoding is already trajectory-exact.*

*Proof.* Write $w_i(\cdot)$ for the common prediction at masked position $i$, which by Definition 5.1 does not depend on the revealed values. The remaining positions are then filled independently from $w$ whatever order the scheduler chooses, so $h_t(x) = \sum_{y \in C} \prod_{i \in \mathcal{M}(x)} w_i(y_i) \prod_{i \notin \mathcal{M}(x)} [y_i = \phi(x_i)]$, which is (2) at $x' = x$, that is $\eta_t(x)$. For the potential, $Z(x'; x)$ and $Z(x'; x')$ clamp the same positions and differ only in which denoiser call supplied the weights at the masked ones; those weights coincide, so the sums are equal and $G_{t+1} = 1$. At the last step $Z(x_T; x_{T-1}) = \mathbf{1}_C(x_T) = \psi_T(x_T)$ and the ratio is again 1. Constant potentials make $\widehat{Z}$ deterministic, and $M$ is then the Doob kernel. □

Two consequences, the second more useful than the first.

A mean-field model does not need particles. The constraint still couples the positions and $p(\cdot \mid C)$ is still not a product, yet one draw from the route samples it exactly. The intuition that "conditioning breaks the product structure, so we need the machinery" is therefore wrong. The dynamic programme handles the coupling. The particles handle something else.

What they handle is re-conditioning, and Proposition 5.2 tells us how to measure it. Define the *re-conditioning strength* as the typical size of

$$\log G_{t+1} = \log Z(x'; x') - \log Z(x'; x),$$

the change in the route's mass caused purely by the model revising its predictions. It is zero for a mean-field model and grows with context sensitivity.

**Consequence for practice.**

The value of this method on a given model scales with that model's re-conditioning strength. A model whose predictions barely move as context arrives is one where step-exact decoding is already nearly unbiased and we add nothing. This is measurable before building anything, by revealing context and watching the predicted distribution at a held-out masked position move (Section 8.5). Reporting that a model sits in the flat regime is as informative as reporting a

gain.

# 6 What goes wrong in practice

Theorem 4.2(iii) says $\mathbb{E}[\widehat{Z}] = h_0$. Unbiasedness is not the same as usefulness: if most of the expectation sits in rare events, the average over any feasible number of runs falls well short. On OWT-130M with TWISTER's own setup, plain resampling gave $\widehat{Z}/h_0 = 0.930$ (CI 0.910–0.949) over $10^5$ runs, while the same configuration with resampling switched off gave 0.998. Resampling, the step meant to help, was losing 7% of the answer. This section explains why and what to do.

## 6.1 Hidden mass

**Lemma 6.1** (Continuation value [ours])**.** *Without further resampling, the expected product of the remaining potentials from state $x$ at step $s$ is $V_s(x) = \mathbb{E}_q[\prod_{t\geq s} G_{t+1} \mid x_s = x] = h_s(x)/\psi_s(x)$.*

*Proof.* From step $s$ on, the proposal kernels and potentials telescope to $p\,\psi_T/\psi_s$ along every path; sum over continuations.

Suppose resampling fires at step $s$. Writing $W_s^k = \rho_s^k\,\psi_s(x_s^k)$ with $\rho_s^k$ the path ratio $p/q$ up to $s$, Lemma 6.1 gives

$$\tilde{w}_k \propto \rho_s^k\,\psi_s(x_s^k), \qquad c_k \propto \rho_s^k\,h_s(x_s^k), \tag{5}$$

where $c_k$ is particle $k$'s share of $\mathbb{E}[\widehat{Z} \mid \mathcal{F}_s]$. There is the mismatch, in one line: **resampling allocates offspring in proportion to $\psi_s$, while each particle's share of the answer is proportional to $h_s$.** Where the frozen twist understates $h_s$, resampling kills particles that were carrying the estimate.

**Proposition 6.2** (Hidden mass under multinomial resampling [ours])**.** *Under multinomial resampling, particle $k$ has at least one offspring with probability $1-(1-\tilde{w}_k)^K \leq K\tilde{w}_k$, and when it does its contribution is multiplied by about $1/(K\tilde{w}_k)$. Define the hidden mass $H_\varepsilon = h_0^{-1}\,\mathbb{E}[\sum_k c_k \mathbf{1}\{\tilde{w}_k < \varepsilon\}]$. Over $R$ runs, the expected number of runs in which any particle with $\tilde{w}_k < \varepsilon$ leaves an offspring is at most $RK\varepsilon\,\mathbb{E}[N_\varepsilon]$. When that is much less than 1, the average of $\widehat{Z}$ over the $R$ runs misses about a fraction $H_\varepsilon$ of $h_0$ with high probability, even though $\mathbb{E}[\widehat{Z}] = h_0$.*

*Proof.* Offspring counts are binomial$(K, \tilde{w}_k)$, giving the first bound; a union bound over particles and runs gives the second. On the event that particle $k$ has $n_k$ offspring they carry weight $\bar{W}$ each, against an expected share $c_k$ that Lemma 6.1 assigns to weight $K\bar{W}\tilde{w}_k$; hence the factor $n_k/(K\tilde{w}_k)$.

Two things follow that are easy to miss. Tail diagnostics fitted to observed values of $\widehat{Z}$, such as the generalised-Pareto shape estimate, *cannot* detect the missing mass, because the events carrying it are absent from the sample. And if $\psi_s = h_s$ then $c_k = \tilde{w}_k \sum_j c_j$ by (5), so $H_\varepsilon \leq K\varepsilon$: hidden mass is a direct measure of twist error. It is large exactly when the frozen twist is *over-confident* that the constraint has become unreachable.

The diagnosis on OWT-130M bears this out. At the first step where resampling would fire, 13% of particles sat below $10^{-6}$ normalised weight while carrying 4.7% of $h_0$, with a median log twist of $-40$; 19% of them recovered above $10^{-3}$ by the end. Those are the particles resampling was discarding.

**Remark 6.3** (Estimating $H_\varepsilon$)**.** Runs with resampling on and off are identically distributed up to the first step resampling would fire. So $H_\varepsilon$ is estimable from resampling-off runs: record every particle's weight at every step, find the first step with ESS $< \tau K$, and sum the final contributions of particles whose normalised weight there was below $\varepsilon$.

**Remark 6.4** (Why synthetic testbeds hide this)**.** $\eta_t$ evaluates the constraint under the current predictions, and nothing forces those to be consistent with what the denoiser will predict later. A real model can assign a completion almost zero probability at one step and substantial probability a few reveals on. A synthetic denoiser whose predictions depend weakly on context has little hidden mass by construction, so tests on such models will not find this.

### 6.2 Three remedies

All three keep $\mathbb{E}[\widehat{Z}] = h_0$ and can be combined. The first two are cheap and come with guarantees independent of the twist's quality; the third attacks the cause.

**Defensive resampling.** Resample ancestors from $\pi = (1-\epsilon)\tilde{w} + \epsilon/K$ instead of $\tilde{w}$, giving each offspring weight $\tilde{w}_a/\pi_a$.

**Proposition 6.5** ([ours])**.** *With this rule:* $\mathbb{E}[\widehat{Z}] = h_0$*; every post-resampling weight is at most* $1/(1-\epsilon)$ *and every particle survives with probability at least* $1 - e^{-\epsilon}$ however small its weight*; and for any non-negative continuation values the resampling step adds a conditional relative variance of at most* $1/\epsilon$*. Under multinomial resampling the corresponding quantity has no bound when a particle with small* $\tilde{w}_k$ *has a large continuation value.*

*Proof.* In Appendix A.

The bound is the point: $1/\epsilon$ holds whatever the twist's errors, and the survival probability $1-e^{-\epsilon}$ does not depend on how badly the twist understates $h_s$. This is defensive importance sampling [9] applied to the ancestor distribution. When the twist is accurate it costs some resampling efficiency; that is the premium. At $\epsilon = 0.25$ it restored $\widehat{Z}/h_0$ from 0.930 to 0.998 on OWT-130M.

**Tempered twists.** Flatten the twist itself by raising it to a power.

**Proposition 6.6** ([ours])**.** *For* $\beta \in (0, 1]$ *let* $\psi_t = \eta_t^\beta$ *for* $t < T$ *and* $\psi_T = \mathbf{1}_C$*, with the step-exact proposal. Then the potential is* $G_{t+1} = \eta_t(x)^{1-\beta}\eta_{t+1}(x')^\beta / Z_d(x'; x)$*, parts (ii) and (iii) of Theorem 4.2 hold, and in* (5)

$$\frac{c_k}{\tilde{w}_k} \;\propto\; h_s(x_s^k)^{1-\beta}\Big(\frac{h_s(x_s^k)}{\eta_s(x_s^k)}\Big)^\beta.$$

*Proof.* In Appendix A.

So an over-confidence factor $\delta^{-1} = h_s/\eta_s$ enters resampling as $\delta^{-\beta}$. The price is that resampling uses only part of the information in the twist; at $\beta \to 0$ particles are resampled on the path ratio alone, and $\beta = 1$ is TWISTER.

**Remark 6.7** (When tempering is not enough)**.** Tempering removes hidden mass only if the tempered mismatch is large enough to be sampled. By Proposition 6.2 a particle suppressed by $\delta^\beta$ leaves an offspring in one of $R$ runs only if $RK\delta^\beta \gtrsim 1$, requiring $\beta \lesssim \log(RK)/\log(1/\delta)$. When the twist is over-confident by many orders of magnitude this forces $\beta$ near 0 and throws away the look-ahead entirely. Defensive resampling has no such condition. It is the safer default; tempering complements it for moderate errors.

**Calibrated twists.** The third remedy changes the law under which the twist is computed, rather than how much of it is used. The frozen twist fills masked positions from the current class weights $w_i$; *any* other product law $u_i$ over the masked positions gives a twist the route still evaluates exactly, at the same cost. The leading choice is $u_i \propto w_i^\tau$ with $\tau < 1$, which flattens the weights.

**Proposition 6.8** (Calibrated twist [ours])**.** *Let $\psi_t = \eta_t^u$ for $t < T$ and $\psi_T = \mathbf{1}_C$, and let $Z_d^{u,r}(x; x)$ be the clamped sum using $w$ on the reveal set $r$ and $u$ elsewhere. Then $M_{t+1}^u(x' \mid x) = s_{t+1}(r \mid x) \prod_{i \in r} p_\theta^i(x'_i \mid x)\, Z_d^u(x'; x)/Z_d^{u,r}(x; x)$ is a probability kernel; it is the step-exact proposal for $\psi = \eta^u$; the potential is $G_{t+1} = \big(Z_d^u(x'; x')/Z_d^u(x; x)\big)\big(Z_d^{u,r}(x; x)/Z_d^u(x'; x)\big)$; and (ii), (iii) of Theorem 4.2 hold. At $u = w$ everything reduces to Lemma 4.1.*

*Proof.* In Appendix A.

**Lemma 6.9** (Calibration floor [ours])**.** *For $u_i \propto w_i^\tau$ with $\tau \in (0, 1]$ and $m = |\mathcal{M}(x)|$, $|\Sigma|^{-m(1-\tau)} \eta_t(x)^\tau \le \eta_t^u(x)$. Hence if the frozen twist is over-confident by a factor $\delta$, the calibrated twist satisfies $h_t/\eta_t^u \le |\Sigma|^{m(1-\tau)} \delta^{-\tau}$.*

*Proof.* In Appendix A.

**Remark 6.10** (What calibration changes, and why the errors are asymmetric)**.** Tempering and calibration both bound over-confidence by a power of $\delta$, but act differently: tempering compresses the total $\eta_t$ and leaves the proposal alone, while calibration compresses each completion's weight inside the route sum, so the proposal *also* looks ahead with $u$ and steers revealed classes towards completions plausible under the flattened law. The asymmetry matters. By (5) a twist that overstates $h_s$ merely wastes allocation, whereas one that understates it by orders of magnitude creates hidden mass. This is the importance-sampling rule that proposals should have heavier tails than the target, applied to the intermediate targets. Nothing requires $\tau \le 1$: for a denoiser whose predictions are too flat, $\tau > 1$ is the corresponding correction.

**Remark 6.11** (Why not an unbiased look-ahead twist?)**.** Proposition 4.5 also permits random twists, for instance $\psi_t = \eta_t \hat{G}_t$ with $\hat{G}_t$ the potential product along a speculative rollout, which is an unbiased estimate of $h_t$ when rolled to step $T$. Unbiasedness is not what resampling needs. By (5) resampling misallocates in proportion to $h_s/\psi_s$, so what matters is the *spread* of $\log(h_s/\psi_s)$ across the particles of one run. Rollout noise adds to that spread, and since $\mathbb{E} \log \hat{G}_t < \log \mathbb{E} \hat{G}_t$, a single rollout typically falls well below its mean when the potential product is heavy-tailed, which understates $h_t$: the harmful direction. A biased twist that varies smoothly across particles can hide less mass than an unbiased noisy one. We measured this and dropped look-ahead twists (Section 8.3).

**Remark 6.12** (Choosing the calibration)**.** $\tau$, a per-step $\tau_t$, or a richer product law can be fitted from a short pilot run by minimising the empirical variance of $\log \widehat{Z}$. This is controlled SMC [8], with the family restricted to product laws so that every route stays exact. A product law cannot represent correlations between future positions, such as those a copy constraint creates; mixtures of product laws, at one route sum per component, are the next family that keeps exactness.

# 7 Ambiguity

By Theorem 4.2 an ambiguous route gives the correct target. It costs variance twice over: the proposal favours completions with many derivations, and the twist $\eta_d = \sum_y d(y) q(y)$ overstates how reachable $C$ is, in proportion to the ambiguity of the remaining completions.

**Proposition 7.1** (Single-step cost of ambiguity [ours]). *Suppose* $1 \le d(y) \le D$ *on* $C$. *Importance sampling from* $q_d \propto d \cdot q \cdot \mathbf{1}_C$ *to* $q(\cdot \mid C)$ *uses weights* $\propto 1/d(y)$, *with ESS fraction at least* $4D/(1+D)^2$, *and the twist ratio obeys* $1 \le \eta_d/\eta \le D$.

*Proof.* Kantorovich's inequality applied to $X = 1/d(y) \in [1/D, 1]$; see Appendix A.

The bound is informative when ambiguity is bounded and vacuous when it is not: for the RNA pairing grammar $D$ grows exponentially in $L$. Early in decoding the twist only sees the ambiguity of completions consistent with the current partial sequence, which can be far smaller than $D$; whether a per-step measure of that kind controls the ESS of whole trajectories is open (Section 10).

The practical consequence is sharper than the bound, and it is a genuine negative result.

**No resampling on ambiguous routes.**

Resampling makes ambiguous routes *worse*, and calibration does not help. The mechanism: the chart's twist is derivation-weighted, so resampling concentrates particles on prefixes with many parse trees rather than high probability. A string-counting twist would fix it, but exact string counting is #P-hard for ambiguous grammars in general, so no exact fix exists. Run ambiguous routes without resampling. Fixed target structures compile to factor graphs, which count each string once, and are unaffected.

Exact counting that ignores derivations is out of reach in general: for CFG constraints on hidden Markov models the exact constrained likelihood is NP-hard already at ambiguity 2, and an FPRAS exists when ambiguity is polynomially bounded [14]. That suggests approximate twists under bounded ambiguity, which Proposition 4.5 permits provided the proposal density is exact and the terminal twist is $\mathbf{1}_C$.

# 8 Experiments

Every number is reported from the saved result file named with it; `docs/status.md` in the repository carries the artifact identifiers. Protocol details are in Appendix C. Three testbeds:

**Synthetic.** A denoiser over $|V| = 40$ symbols whose context sensitivity is set by a scale parameter, so the exact conditional can be enumerated and the sampler checked against it.

**OWT-130M.** The released MDLM text model of [19], the setting of TWISTER's own experiments.

**mRNAutilus.** A 150M-parameter masked discrete diffusion model over full-length mRNA, used here only as the subject of a model diagnostic; the design experiments are future work (Section 9).

## 8.1 Six constraint families on a real model

Families chosen to exercise different routes: COPY (factor graph), REGULAR and COUNTING (automata), DYCK1-UNAMB and DYCK1-AMB (charts, one ambiguous), and RNA-PAIR (the Nussinov–Jacobson pairing grammar [15], ambiguous). OWT-130M, $L = 32$, $T = 16$, $K = 16$. The metric is pairwise total variation against a rejection reference in excess of a matched sampling floor, so 0 is as good as the reference; bootstrap standard errors on the excess are 0.003–0.008.

| | step-exact | no resampling | | defensive $\epsilon = 0.25$ | | plain |
|---|---|---|---|---|---|---|
| family | $K = 1$ | $\tau = 1$ | $\tau = 0.5$ | $\tau = 1$ | $\tau = 0.5$ | $\tau = 1$ |
| copy | 0.165 | 0.136 | 0.068 | 0.148 | 0.040 | 0.135 |
| regular | 0.129 | 0.026 | 0.046 | 0.061 | 0.057 | 0.089 |
| counting | $\approx 0$ | $\approx 0$ | $\approx 0$ | $\approx 0$ | $\approx 0$ | $\approx 0$ |
| dyck1-unamb | 0.160 | 0.133 | n/a | 0.132 | n/a | 0.132 |
| dyck1-amb | 0.154 | 0.086 | 0.091 | 0.143 | 0.158 | 0.151 |
| rna-pair | 0.030 | 0.005 | 0.006 | 0.172 | 0.140 | 0.212 |

Three readings. Step-exact decoding carries substantial excess on five of six families, the bias Proposition 5.2 says must come from re-conditioning. Particles remove most of it. And resampling *hurts* on the two ambiguous grammars, as Section 7 predicts. The tail diagnostic agrees: the Pareto shape $\hat{k}$ on COPY falls from 1.07 (step-exact) to 0.37 ($K = 16$) to 0.08 with calibration.

### 8.2 Long horizons

The same families at $L = 64$, $T = 64$, $K = 8$, quadrupling the weight increments per trajectory. COPY behaves as above (TV excess 0.095 without resampling, 0.038 with defensive resampling and $\tau = 0.5$), though that best arm has $\widehat{Z}/h_0 = 0.902$ (CI 0.827–0.983, $\hat{k} = 0.67$): the best samples with a visibly biased normalising constant. COUNTING is exact on every arm. RNA-PAIR is the clearest remaining failure: defensive resampling gives TV excess 0.043 but $\widehat{Z}/h_0 = 0.576$ (CI 0.359–0.858, $\hat{k} = 1.00$), an infinite-mean weight tail.

### 8.3 The calibrated twist, and the look-ahead twists we dropped

Synthetic testbed, context scale 8, $K = 4$, $2 \times 10^4$ runs per arm, sweeping $\tau$. Efficiency is the ratio of $\widehat{Z}$ variance to the resampling-off arm at $\tau = 1$, so 1 means the resampling penalty is gone.

| $\tau$ | no resampling | plain | defensive $\epsilon = 0.25$ | hidden mass $< 10^{-2}$ | resamplings/run |
|---|---|---|---|---|---|
| 1 | 1.00 | 0.14 | 0.47 | 7.8% | 1.78 |
| 0.7 | 0.99 | 0.32 | 0.78 | 3.2% | 1.69 |
| 0.5 | 1.02 | 0.63 | 1.09 | 0.9% | 1.56 |
| 0.35 | 1.01 | 1.04 | 1.15 | 0.16% | 1.40 |
| 0.2 | 1.03 | 1.09 | 1.04 | 0.15% | 1.17 |

All fifteen arms give $\widehat{Z}/h_0$ within 0.979–1.008, as Proposition 6.8 requires: calibration changes the variance, never the target. On the resampling-on arms the mechanism is the one Lemma 6.9 describes, and the hidden-mass column tracks it directly.

Look-ahead twists were measured in the same testbed and abandoned. Plain resampling with a one-step look-ahead reached efficiency 0.001, and with full look-ahead likewise; the fraction of $h_0$ hidden below $10^{-6}$ rose from 0.0004 (frozen twist) to 0.069 (full look-ahead). Geometric averaging over rollouts cut that to 0.006, still 20 times the frozen twist. This is the asymmetry of the previous section in numbers: the rollout noise widens the spread of $\log(h_s/\psi_s)$ more than the bias it removes.

**Remark 8.1** (A claim we withdrew)**.** We previously attributed calibration's benefit with resampling *off* to smoother per-step weight increments. That is false. A $K = 1$ trace decomposition shows summed per-step log-weight variance falling $11.1 \to 6.0$ as $\tau$ goes $1 \to 0.5$, while total path log-weight standard deviation barely moves ($3.278 \to 3.298$) and the relative standard error of $\widehat{Z}$ is flat ($0.0128 \to 0.0132$). Across families at $K = 16$ with resampling off, the ratio of relative standard

errors ($\tau = 0.5$ vs $\tau = 1$) is 1.02, 1.17, 0.30 and 0.66 on PREFIX, COPY, COUNTING and FORBIDDEN: family-dependent, sometimes worse, and not explained by increment smoothness. The effect is real where it appears; its mechanism is open.

### 8.4 Ambiguity

Synthetic testbed with grammar routes, verifying Theorem 4.2: at $L = 4$ and $L = 6$ the sampled law matches the exact conditional to total variation $\leq 4 \times 10^{-16}$, ambiguity notwithstanding. The cost is in the weights. At $K = 16$ the ESS fraction over $L = 8$ to 32 holds at 0.93–0.97 for unambiguous Dyck-1 but falls from 0.92 to 0.34 for the ambiguous grammar and from 0.89 to 0.30 for RNA-PAIR, tracking the mean log derivation count, which reaches 4.4 and 5.3.

### 8.5 Is there anything to correct on an mRNA model?

A direct measurement of re-conditioning strength (Section 5) before committing to the application. Holding one codon position masked and revealing the rest of a coding sequence in four stages, the predicted distribution at the held-out position moves by total variation 0.337 end to end, or 0.205 restricted to the synonymous codons of the true residue, with the largest single movement, 0.291, in the first quarter of reveals, which is where step-exact decoding commits. A trajectory mean-field model would give 0 at every stage. Two related measurements: randomising the whole context moves the prediction by 0.336, and a synonymous shuffle preserving the encoded protein still moves it by 0.261. The model's joint is not a product of per-position codon tables.

Its *mode*, however, is. On held-out synonymous-codon prediction ($n = 548$ paired positions) the model is top-1 correct 39.8% of the time against 38.7% for a human codon-frequency lookup table and 29.5% for uniform choice among synonyms; McNemar's test gives $z = +0.53$, not significant. These findings are compatible and say different things: a product model reproduces the marginal mode and cannot reproduce the joint.

Caveats: one coding sequence, which appears in the model's own configuration and may be in its training data; the perturbed contexts are somewhat out of distribution; $n = 6$ held-out positions for the re-conditioning figures. These are order-of-magnitude indications and should be repeated across several real human coding sequences.

## 9 Application: mRNA sequence design

### 9.1 The design problem

A protein is fixed, since it is the product, but the genetic code is degenerate, so most amino acids have two to six synonymous codons and a 300-residue protein has on the order of $10^{150}$ coding sequences. They differ in translation efficiency, stability, folding near the start codon, immunogenicity, and manufacturability: whether a restriction site breaks the cloning step, whether a homopolymer run causes synthesis errors. Choosing well within that equivalence class is codon optimisation.

What makes this a good testbed, beyond the application itself, is that mRNA design has something most generation tasks lack: a deterministic, cheap, non-learned scorer. ViennaRNA computes exact minimum free energy and ensemble defect. Candidates can be evaluated without trusting a learned reward model, which is the usual weak point of controllable-generation evaluations.

### 9.2 How the system is layered

Three layers, with strict dependence in one direction. *Layer 1* is everything in Sections 3, 4 and 6 and knows nothing about biology. *Layer 2* is the codon alphabet, the genetic code, and constraint builders that return layer-1 route objects over an abstract alphabet. *Layer 3* is the model adapter.

Protein identity enters *through the route*, as per-position allowed-codon masks, never by renormalising the model's predictions. This is the methodological point of the whole application: mRNAutilus's own conditional sampler imposes protein identity by masking logits at each step, which is exactly the step-local projection that Example 1.1 shows to be biased. Our route imposes the same requirement without touching the trajectory law.

Constraints compile as expected. For a 60-residue protein: motifs alone give a 20-state automaton; motifs plus a GC band give 2,174 states in 1.0 s; motifs, GC band and a 7-base-pair hairpin give 968 states, domain 213, induced width 3, in 0.5 s. The automaton-times-structure product initially needed $1792^3$ cells and now enumerates only feasible (codon, state) pairs.

### 9.3 What to measure, and what not to

A survey of the codon-optimisation literature settled a question that changes what we claim. **No published learned codon-optimisation method reports per-position codon reconstruction accuracy against a frequency baseline as a primary result.** The single instance in the literature is an unnumbered density plot at nucleotide resolution in which the frequency baseline is barely separated from uniform random synonymous choice. The field reports distributional agreement with held-out natural sequences, positional rare-codon placement, and downstream property prediction instead. Our flat top-1 result in Section 8.5 is therefore not a failure to reproduce published gains; it is a metric those gains were never claimed on. It stays as a one-line sanity check that the model is not broken, which is what it established.

The constrained families are the headline, because that is where rule-based construction fails outright rather than merely underperforming: target secondary structure, GC content in sliding windows, motif avoidance where the forbidden motif straddles a codon boundary, and conjunctions of these. A frequency table cannot satisfy any of them; a greedy repair heuristic satisfies them while distorting the distribution in ways we can measure exactly.

Planned measurements, in order of how directly they test our claim:

1. *Exactness and coverage.* The fraction of generated sequences satisfying the amino-acid constraint exactly (1 by construction for us, measurably less for step-local projection), and the total variation between our sampler's law over the synonymous lattice and the model's true conditional, on short coding sequences where the lattice can be enumerated. This is the claim no incumbent can state.
2. *Positional profile against three controls.* CAI, length-normalised MFE, GC and GC3, and a positional rare-codon profile distance to the natural sequence, against a frequency-table draw, a max-CAI greedy construction, and the natural sequence. The positional term is the discriminating one: it rewards reproducing *where* rare codons sit, which a product model is structurally incapable of.
3. *Best-of-$K$ under a non-learned scorer*, with diversity at fixed $K$ reported alongside, against mRNAutilus's own mode-seeking sampler.

### 9.4 Why a diverse library, specifically

Three reasons that stack. Every computational mRNA score correlates imperfectly with what happens in cells, so taking the argmax under an imperfect score bets everything on its top-ranked

point, whereas synthesising many diverse high-scoring candidates samples the score's upper region. Optimisers with restarts give near-duplicates around one mode, while a calibrated sampler gives spread at controlled quality. And every candidate must be manufacturable, so a library where 30% of members violate a constraint wastes 30% of the synthesis budget. Hard guarantees plus diversity is the combination existing methods do not offer together: optimisers give quality without diversity, unconstrained samplers give diversity without guarantees, and step-local projection gives guarantees with a silently distorted distribution.

### 9.5 State of the implementation

The model adapter is written and the encoder validated: the vendored transformer block is bit-identical to upstream's on identical inputs, after fixing a reversed SwiGLU gate order and a missing token-dropout rescaling. An end-to-end smoke test samples constrained coding sequences, with all particles live, every sequence translating to the target peptide, and none containing a forbidden restriction site. The design experiments have not been run.

## 10 Open directions

### 10.1 Can the twist be made less over-confident without adding noise?

Proposition 6.2 makes over-confidence of the frozen twist the main practical limit, and Proposition 4.5 leaves the intermediate twists entirely free. Calibration (Proposition 6.8) helps and look-ahead (Section 8.3) does not, for the asymmetry reason. What is unresolved is whether a twist exists that is both smooth across particles and genuinely more accurate: candidates are per-step $\tau_t$ fitted from a pilot run, mixtures of product laws for correlated constraints at one route sum per component, and learned twists $\psi_\vartheta \approx h_t$ fitted to native rollouts as in [25], with $\eta_t$ as a feature.

### 10.2 Why does calibration help with resampling off?

It does, on some families (Section 8.3), and our first explanation was wrong. The effect is family-dependent (a 3.3× improvement on COUNTING, a 1.17× degradation on COPY) and is not mediated by smoother per-step increments. Isolating the real mechanism would probably tell us something about which constraint geometries the calibrated proposal suits.

### 10.3 Can hidden mass be bounded online?

Proposition 6.2 expresses hidden mass through $h_s/\psi_s$, which cannot be observed, and tail diagnostics provably cannot see it. An estimate or upper bound computable *during* a run, perhaps from a small population of particles that is never resampled, would turn an offline diagnostic into a live check. This is the single most useful missing piece for anyone deploying the method.

### 10.4 Does bounded per-step ambiguity control trajectory ESS?

Proposition 7.1 bounds the single-step cost by the global derivation range $D$, which is vacuous when $D$ grows exponentially. Early in decoding the twist only sees the ambiguity of completions consistent with the current partial sequence, which is far smaller. A bound on whole-path ESS in terms of such a local measure would make the ambiguous case predictable rather than merely survivable.

### 10.5 How far can superset routes be pushed?

For a context-free $C$, propose from a regular superset $R \supseteq C$ with twist $\psi_t = Z_R(x; x) \cdot \mathrm{comp}_C(x)$, where $\mathrm{comp}_C$ tests completability. By Proposition 4.5 this is exact provided $C \subseteq R$ and $\mathrm{comp}_C$ never rejects a completable state. Reporting 1 for a dead state only wastes a particle, while a regular *subset* is biased unless the bound provably covers every string of length $L$. Pruning converts one accept-or-reject at the end into deaths at individual steps, which resampling can repair, provided $R$ is tight enough to keep the per-step death rate well below $1 - 1/K$. A [proposed] variant: rather than letting a dead proposal kill its particle, redraw until $\mathrm{comp}_C$ accepts, and use an unbiased estimate of the single-draw acceptance probability in the potential, which is valid by the random-twist part of Proposition 4.5.

### 10.6 Where does exact sampling overtake mode-seeking in best-of-$K$?

Exact conditional sampling should eventually beat mode-seeking decoding as $K$ grows, since the mode is a single point and the sampler explores the upper region of the score. Where the crossover lies, and how it depends on the scorer's correlation with the true objective, is open, and is what the mRNA best-of-$K$ experiment (Section 9) is designed to measure.

### 10.7 Beyond a fixed protein

The route machinery is indifferent to biology. The natural next targets are better cases for the method than coding-sequence design: 5′UTR design has no protein constraint at all, so the model's joint carries all the information, and translation initiation depends on structure near the cap and start codon: global constraints that rules handle badly. Constraints spanning the UTR–CDS boundary are where codon-table methods are structurally incapable rather than merely worse. Structured text generation is the same algorithm with a different compile target.

## 11 Related work, by component

**The trajectory correction.** TWISTER [22] is the direct predecessor: twisted SMC for masked diffusion with DFA constraints, including the step-exact proposal (Lemma 4.1) and the exactness theorem for that case. Zhao et al. [25] apply twisted SMC to autoregressive language models with learned twists. Dang et al. [4] observe that a derivation-weighted NFA proposal remains valid in autoregressive SMC. The underlying Feynman–Kac theory, including unbiasedness of $\widehat{Z}$ under adaptive resampling, is [6].

**Step-local constrained decoding.** Dang and Ermon [5] give the automaton-based step-exact decoder that Example 1.1 shows to be biased across steps; it is our $K = 1$ arm. Su and Zhang [21] make the case that constraints are graphs rather than chains, which is the factor-graph route here.

**What is tractable in one step.** The circuit characterisation is Broadrick et al. [3]; the semiring behaviour under ambiguity is a special case of the compositional atlas of Wang et al. [23]. Jerrum, Valiant and Vazirani [11] give the sampling-to-counting reduction the compiler uses to decide refusals, and Jerrum and Snir [10] the monotone lower bound that forces the determinant route to exist. Marzouk and de la Higuera [14] settle the CFG-constrained case under bounded ambiguity.

**The individual routes.** Grammar charts and top-down sampling are classical; the RNA pairing grammar is Nussinov and Jacobson [15]. The determinant route uses the matrix-tree marginals of Koo et al. [13] and Wilson's sampler [24].

**The remedies.** Defensive resampling adapts defensive importance sampling [9] to the ancestor distribution. Calibrated twists are a restricted, closed-form member of the controlled-SMC family [8], which fits twists by regression within a parametric family; restricting to product laws is what keeps every route exact.

**The model.** Masked diffusion language models follow Sahoo et al. [19], whose reverse step defines the independent schedule of Section 2.

**Who else composes constraints of this kind.** Two axes separate the prior work: which constraint classes a method can express, and which semiring it evaluates them in. On expressiveness, no decoder matches the set in Table 1: TWISTER [22] is trajectory-exact but automaton-only; Su and Zhang [21] argue for graphs over chains and give the factor-graph case; grammar-constrained decoding [7] handles context-free constraints for autoregressive models by per-step masking, which is biased in the same way Example 1.1 describes, and grammar-aligned decoding [18] removes that bias by iteratively reweighting across samples rather than within a trajectory.

The closest prior construction is not in the decoding literature at all. LinearDesign [20] encodes the synonymous-codon design space of a protein as a weighted DFA and runs *lattice parsing* against an RNA folding grammar — the automaton-times-grammar conjunction of Appendix B, built for exactly our application. The difference is the semiring. LinearDesign evaluates the intersection in max-times and returns the single optimum of a hand-written objective (free energy traded against codon adaptation); we evaluate it in sum-product and return samples from a learned model's conditional. By Proposition 3.2 that difference is not cosmetic: the max semiring is correct whatever the grammar's ambiguity, while the sum semiring counts each string once per derivation. LinearDesign can intersect with an ambiguous folding grammar and ignore the issue; we cannot, which is what Section 7 is about. So the honest statement of novelty is narrow and worth stating precisely: the conjunction is not new, and performing it in the sum semiring over a learned distribution is.

## 12 Summary

**Where this stands.**

The trajectory-level correction for masked diffusion decoding is independent of the constraint representation: any route with a matching mass and sampler, plus an exact terminal membership test, gives exact conditional sampling (Theorem 4.2), ambiguous grammars included. The method's value on a given model is governed by that model's re-conditioning strength (Proposition 5.2), which is measurable in advance. Its main practical failure mode is hidden mass under resampling (Proposition 6.2), for which defensive resampling gives an unconditional guarantee and calibration an improvement that depends on the constraint family. Ambiguous routes should not be resampled, and this admits no exact fix.

| result | status | source or note |
|---|---|---|
| Semirings and ambiguity (Proposition 3.2); MAP for any CFG | background | [23] |
| Circuit characterisation (Theorem 3.3) | background | [3, 11] |
| Spanning trees separate monotone from general circuits (Remark 3.4) | background | [10] |
| Step-exact proposal (Lemma 4.1); free twists (Proposition 4.5) | background | [22, 6] |
| Routes with single-pass exact samplers; compiler with refusal | ours (system) | Section 3 |
| Route-generic trajectory exactness (Theorem 4.2) | ours | extends [22]; compare [4] |
| Terminal-twist pitfall (Proposition 4.3) | ours | |
| Mean-field collapse (Proposition 5.2); re-conditioning strength | ours | delimits when the method is needed |
| Two-sided identity (Lemma 4.6) | ours | diagnostic |
| Continuation value and hidden mass (Lemma 6.1 and Proposition 6.2) | ours | measured at 7% of $h_0$ |
| Defensive resampling (Proposition 6.5) | ours | compare [9] |
| Tempered intermediate twists (Proposition 6.6) | ours | |
| Calibrated twists and their floor (Proposition 6.8 and Lemma 6.9) | ours | compare [8] |
| Single-step cost of ambiguity (Proposition 7.1); no resampling rule | ours | no exact fix exists |
| Look-ahead twists measured and dropped | ours | Section 8.3 |
| Superset routes with Boolean pruning | ours | Section 10 |
| Online hidden-mass guard; trajectory ESS under ambiguity | proposed | Section 10 |

# A Proofs

*Proof of Theorem 4.2.* (i) By Lemma 4.1, $q = \prod_t \kappa_{t+1} Z_d(x_{t+1}; x_t)/Z_d(x_t; x_t)$. Multiplying by (4), the $Z_d$ factors cancel and the twists telescope to $qW = \psi_0 \prod_t \kappa_{t+1}\psi_{t+1}/\psi_t = p\,\psi_T = p\,\mathbf{1}_C$. The divisions are legitimate on the relevant paths: if $x'$ can still be completed to an element of $C$ then some $y \in C$ agrees with $x'$, so $d(y) > 0$ and hence $Z_d(x'; x) > 0$ under the positivity assumption, giving $M_{t+1}(x' \mid x) > 0$. If $\psi_t(x) = 0$ the particle already has weight zero and we set $0/0 = 0$.

(ii) follows from (i) by the standard Feynman–Kac argument [6].

(iii) Unbiasedness under multinomial resampling with adaptive decisions is standard [6]. For $\epsilon > 0$, let $\mathcal{F}$ be the particle system just before a resampling step, with weights $W^k$ and mean $\bar{W}$. For any $f$,

$$\mathbb{E}\Big[\bar{W}\,\frac{1}{K}\sum_j \frac{\tilde{w}_{a_j}}{\pi_{a_j}} f(x^{a_j}) \;\Big|\; \mathcal{F}\Big] = \bar{W}\sum_k \tilde{w}_k f(x^k) = \frac{1}{K}\sum_k W^k f(x^k),$$

because $\pi_k > 0$ wherever $\tilde{w}_k > 0$. A resampling step therefore leaves the conditional expectation of the weighted particle measure unchanged, whether or not it fires. Iterating gives $\mathbb{E}[\widehat{Z}] = \mathbb{E}_q[W] = h_0$ by (i).

(iv) At the last step $x_T$ is fully revealed, so $Z_d(x_T; x_{T-1}) = d(x_T)$, and (4) gives $G_T =$

$Z_d(x_{T-1}; x_{T-1})\mathbf{1}_C(x_T)/\big(\psi_{T-1}(x_{T-1})\, d(x_T)\big)$: the derivation count enters the denominator and cancels the one the proposal introduced. □

*Proof of Proposition 6.5.* (i) is Theorem 4.2(iii). (ii) follows from $\pi_k \geq (1-\epsilon)\tilde{w}_k$ and $\pi_k \geq \epsilon/K$. For (iii), let $Y = (\tilde{w}_a/\pi_a)\, g_a$ with $a \sim \pi$. Then $\mathbb{E}Y = \sum_k \tilde{w}_k g_k$ and $\mathbb{E}Y^2 = \sum_k \tilde{w}_k^2 g_k^2/\pi_k \leq (K/\epsilon)\sum_k (\tilde{w}_k g_k)^2$. The variance of an average of $K$ independent copies is at most $\mathbb{E}Y^2/K$, and $\sum_k (\tilde{w}_k g_k)^2 \leq (\sum_k \tilde{w}_k g_k)^2$ because $g \geq 0$. Dividing gives the bound $1/\epsilon$. Under multinomial resampling the same ratio is $\frac{1}{K}\big(\sum_k \tilde{w}_k g_k^2/(\sum_k \tilde{w}_k g_k)^2 - 1\big)$, unbounded when a particle with small $\tilde{w}_k$ has large $g_k$. Taking $g_k = V_s(x^k)$ from Lemma 6.1 covers the situation of Proposition 6.2. □

*Proof of Proposition 6.6.* $\eta_t^\beta > 0$ wherever $\eta_t > 0$, which includes every completable state, so Proposition 4.5 applies. The potential is $G_{t+1} = \eta_t(x)^{1-\beta}\eta_{t+1}(x')^\beta / Z_d(x'; x)$, and the expression for $c_k/\tilde{w}_k$ is (5) with $\psi_s = \eta_s^\beta$. □

*Proof of Proposition 6.8.* (i) Fix $r$. $Z_d^u(x'; x)$ depends on the tokens revealed on $r$ only through their classes, so summing $\prod_{i\in r} p_\theta^i(x_i' \mid x)\, Z_d^u(x'; x)$ over tokens on $r$ gives $\sum_c \prod_{i\in r} w_i(c_i \mid x)\, Z_d^u(x_c'; x) = Z_d^{u,r}(x; x)$, where $x_c'$ is $x$ with classes $c$ on $r$. The marginal on $r$ of the measure SAMPLE draws from is $\prod_{i\in r} w_i(c_i \mid x)\, Z_d^u(x_c'; x)/Z_d^{u,r}(x; x)$, and tokens within classes contribute $p_\theta^i(x_i' \mid x)/w_i(c_i \mid x)$. (ii) is the form of $M^u$. (iii) is (3) with $\kappa/M^u = Z_d^{u,r}(x; x)/Z_d^u(x'; x)$. (iv) $u_i > 0$ wherever $w_i > 0$, so $\eta_t^u > 0$ wherever $\eta_t > 0$, which includes every state with $h_t > 0$, and $M^u > 0$ wherever $\kappa h > 0$; Proposition 4.5 applies. □

*Proof of Lemma 6.9.* Since $\sum_\rho w_i(\rho) = 1$ and $0 < \tau \leq 1$, Hölder gives $1 \leq \sum_\rho w_i(\rho)^\tau \leq |\Sigma|^{1-\tau}$, so $|\Sigma|^{-(1-\tau)} w_i^\tau \leq u_i \leq w_i^\tau$ at every masked position. Multiply over positions and sum against $d$. For the lower bound use $(\sum_j b_j)^\tau \leq \sum_j b_j^\tau$, counting each derivation of $y$ as one term. The last claim uses $h_t \leq 1$. □

*Proof of Proposition 7.1.* Let $X = 1/d(y)$ with $y \sim q_d$, so $X \in [m, M]$ with $m = 1/D$, $M = 1$, and write $\mu = \mathbb{E}X$. From $(X - m)(M - X) \geq 0$ we get $\mathbb{E}X^2 \leq (m + M)\mu - mM$. The quadratic $\frac{(m+M)^2}{4mM}\mu^2 - (m+M)\mu + mM$ has zero discriminant, hence is non-negative, giving $\mathbb{E}X^2 \leq \frac{(m+M)^2}{4mM}\mu^2$ (Kantorovich). Substituting $m = 1/D$, $M = 1$ gives the ESS bound; the twist bound is immediate from $1 \leq d \leq D$. □

# B Route implementations

**Automata.** For $(Q, q_0, \delta, F)$ over $\Sigma$, the forward recursion $\alpha_0(q) = [q = q_0]$, $\alpha_i(q) = \sum_{q', \sigma: q \in \delta(q', \sigma)} \alpha_{i-1}(q') w_{i,\sigma}$, MASS $= \sum_{q\in F} \alpha_L(q)$ costs $O(L|\delta|)$. SAMPLE is forward filtering, backward sampling: draw $q_L \propto \alpha_L(q)[q \in F]$, then for $i = L, \ldots, 1$ draw $(q_{i-1}, \sigma_i)$ with probability $\propto \alpha_{i-1}(q_{i-1}) w_{i,\sigma_i} [q_i \in \delta(q_{i-1}, \sigma_i)]$. For a DFA $d = \mathbf{1}_C$; for an NFA $d(y)$ counts accepting paths. Compiles: forbidden and required motifs via Aho–Corasick, counting conditions ($O(Lk)$ states), regular expressions over classes, bounded bracket depth, sliding-window all-different, and products of these. The forward pass must be in float64. In float32 it returned $Z = 0$ for a feasible constraint at log-mass $-113.5$.

**Grammar charts.** For a grammar in Chomsky normal form, the inside algorithm

$$\beta(A, i, i{+}1) = \sum_\sigma [A \to \sigma] w_{i,\sigma}, \qquad \beta(A, i, j) = \sum_{A \to BC} \sum_{i<k<j} \beta(B, i, k)\beta(C, k, j), \qquad \text{MASS} = \beta(S, 0, L)$$

costs $O(L^3|G|)$; leaves for all positions come from one matrix product of the denoiser output with the lexical map. SAMPLE pushes $(S, 0, L)$ and repeatedly pops $(A, i, j)$: at a leaf draw $\sigma \propto [A \to \sigma] w_{i,\sigma}$, otherwise draw $(A \to BC, k) \propto \beta(B, i, k)\beta(C, k, j)$ and push both children. This draws each parse tree with probability proportional to its weight, so the sampled string has probability $\propto d(y) \prod_i w_{i,y_i}$. On a GPU all spans of one length are processed together. Per-nonterminal span limits (a subtree rooted at a listed nonterminal covers at most $W$ positions) cut the work to $O(L^2 W|G|)$ when every nonterminal outside a right-branching spine is bounded.

**Factor graphs.** With 0/1 factors and an elimination order of induced width $\omega$, variable elimination gives MASS in $O(L|\Sigma|^{\omega+1})$, and SAMPLE eliminates in that order and samples in reverse, each bucket table being the exact conditional. Derivations are assignments, so $d = \mathbf{1}_C$, which is why fixed target structures are immune to the ambiguity problem. The motivating case is the same-order copy relation [21], of width 1: MASS $= \prod_{i \le k} \sum_\sigma w_{i,\sigma} w_{i+k,\sigma} \times \prod_{i > 2k} \sum_\sigma w_{i,\sigma}$. Elimination cost is governed by the induced width of the order, and the smallest width over all orders is the treewidth; finding an optimal order is NP-hard [2], so we use the greedy minimum-degree heuristic (George and Liu, SIAM Review 31(1):1–19, 1989).

**Determinants.** Let $w_m(h)$ be the weight that node $m$ has head $h$, and $\hat{L}$ the reduced Laplacian with $\hat{L}_{mm} = \sum_{h \ne m} w_m(h)$ and $\hat{L}_{hm} = -w_m(h)$. The directed matrix-tree theorem gives MASS $= \det \hat{L}$, and differentiating $\log\det \hat{L}$ gives the marginals $\mu_m(h) = w_m(h)([\hat{L}^{-1}]_{mm} - [h \ne 0][\hat{L}^{-1}]_{mh})$ [13]; both cost $O(n^3)$. SAMPLE is Wilson's algorithm [24]: for each node not yet in the tree, run the random walk moving $m \to h$ with probability $w_m(h) / \sum_{h'} w_m(h')$ until it meets the tree, erase loops, and add the path.

**From classes to tokens.** A route draws a class sequence $y$; each revealed position then draws a token within its class, $v \propto p^i_\theta(v \mid x)[\phi(v) = y_i]$. The combined token-level law is $\prod_i p^i_\theta(v_i \mid x)\mathbf{1}_C(\phi(v))/Z$, which is what the trajectory layer requires. Class masses must be computed by exact per-class log-sum-exp rather than a floored sum: flooring at $10^{-38}$ inflates rare classes and biases the weights precisely in the regime that matters.

**Conjunctions.** Automaton with automaton gives a product automaton. Automaton with grammar gives a grammar, by the Bar-Hillel triple construction, of size at most $|R|Q^3$ and usually far less. Automaton with factor graph gives a factor graph over augmented variables $z_i = (y_i, s_i)$, with the automaton contributing a chain of pairwise factors; restricting to feasible (symbol, state) pairs per position is what makes this practical. Two grammars, or a grammar with a factor graph, are refused, as is any conjunction whose counting problem is #P-hard.

# C Experimental protocol

**Estimators and diagnostics.** $\widehat{Z}$ ratios are reported against $h_0$ estimated from a large native sample, with bootstrap confidence intervals that include the binomial uncertainty in $h_0$. $\hat{k}$ is the generalised-Pareto shape of the upper weight tail, fitted as in Pareto-smoothed importance sampling with the Zhang–Stephens estimator; $\hat{k} > 0.5$ indicates infinite variance, $> 0.7$ that the plain mean is unreliable, $> 1$ infinite mean. Sample quality is pairwise total variation between empirical joint class distributions of every position pair, averaged over pairs, measured against a rejection reference and reported in excess of a matched sampling floor.

**Correctness checks.** Each route is verified against brute-force enumeration at small $L$; the sampler is verified for $\mathbb{E}[\widehat{Z}] = h_0$; golden regression values pin the RNG stream so that refactors cannot silently change results. Four correctness bugs were found this way, each a silent failure rather than a crash: a float32 automaton forward pass returning $Z = 0$ at log-mass $-113.5$; class probabilities floored at $10^{-38}$ biasing weights in the rare-event regime; padded columns counted as reveals under the independent schedule, biasing $\widehat{Z}$ by about 30%; and a scheduler refactor that moved the reveal draw ahead of the route sample, changing the RNG order.

**Reproduction.** Result files and artifact identifiers are listed in `docs/status.md`. The test suite covers compilation, CPU and batched samplers, SMC, resampling, diagnostics, exact laws, alphabets, and the biology layers.

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